

CH1 Axioms of Probability

1.2 Sample space and events

Definition

Experiment: 一個機率問題是由一個機率事件 (e.g. tossing a die)

Outcome: 產生結果 (A.K.A sample point)

Sample space: 所有可能的結果集合 (e.g. 丟硬幣 {Head, Tail})

Event: 所有可能的子集合 (e.g. 硬幣頭在上的可能性 {Head})

Ex 1.3 (An infinite sample space)

Measuring the lifetime of a light bulb, where last at least 100 hrs

$$S = \{x : x \geq 0\} \quad E = \{x : x \geq 100\}$$

Ex 1.4 (An finite sample space) (a.k.a Classical Probability)

All families with 1, 2, or 3 children (genders specified)

$$S = \{b, g, bg, gb, bb, gg, bbb, bbg, bgg, gbg, ggb, gbb\}$$

Definition

Occur: Event E has occurred if result contains a element in E

$E \subset F$ E is a subset of F

$E \cap F = EF$ Intersection of E and F

$E \cup F$ Union of E and F

$E - F$ The set difference of E and F

$E^c = \bar{E}$ All element excluding E = $S - E$

$$\cup_{i=1}^n E_i \quad E_1 \cup E_2 \cup E_3 \dots E_n$$

$$\cap_{i=1}^n E_i \quad E_1 \cap E_2 \cap E_3 \cap \dots \cap E_n$$

$$\cup_{i=1}^{\infty} E_i, \cap_{i=1}^{\infty} E_i$$

Sample space and events

Commutative laws (交換律) $E \cup F = F \cup E$, $E \cap F = F \cap E$

Associative laws (結合律) $E \cup (F \cap G) = (E \cup F) \cap G$

Distributive laws (分配律) $(E \cup F) \cap H = (E \cap H) \cup (F \cap H)$

$$(E \cap F) \cup H = (E \cup H) \cap (F \cup H)$$

De Morgan's law 1st rule $(E \cup F)^c = E^c \cap F^c$

2nd rule $(E \cap F)^c = E^c \cup F^c$

E and F are **mutually exclusive** if they have nothing in common $E \cap F = \emptyset$

無中生有 $E = E \cap S = E \cap (F \cup F^c) = EF \cup EF^c$ (也是因為 EF 與 EF^c 是 mutually exclusive)

1.3 Axioms of probability

S (sample space): 所有的可能性的集合

A (event): 其中可能性的集合

Pr (Function): 發生 A 事件的機率為 P(A)

Axioms 1: $P(A) \geq 0$ 機率必定大於 0

Axioms 2: $P(S) \leq 1$ 機率最大為 1

Axioms 3: 只有在所有 event 皆 mutually exclusive 時, 機率才為每個 event 的和

If $\{A_1, A_2, A_3, \dots, A_n\}$ are mutually exclusive then: $P(\cup_{i=1}^{\infty} A_i) = \sum_{i=1}^{\infty} P(A_i)$

Classical definition of Probability

1. Sample space is finite

2. Each event is **equally likely** to occur (shares a same probability)

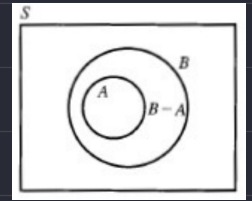
$P(A) = N(A)/N(S)$... N(A) is the number of sample points of A.

1.4 Basic Theorem

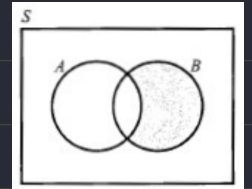
Theorem 1.4: $P(A^c) = 1 - P(A)$

Theorem 1.5: If $A \subset B$, then $P(B - A) = P(BA^c) = P(B) - P(A)$

Corollary If $A \subset B$, then $P(A) \leq P(B)$



Theorem 1.6: $P(A \cup B) = P(A) + P(B) - P(AB)$



■ Inclusion-Exclusion Principle

$$P(A_1 \cup A_2 \cup \dots \cup A_n) = \sum P(A_i) - \sum P(A_i A_j) + \sum P(A_i A_j A_k) - \dots + (-1)^{n-1} P(A_1 A_2 \dots A_n)$$

- **Recall the continuity of a function $f: \mathbb{R} \rightarrow \mathbb{R}$**

$$\lim_{n \rightarrow \infty} f(x_n) = f\left(\lim_{n \rightarrow \infty} x_n\right)$$

for every convergent seq $\{x_n\}$ in $\mathbb{R}$.

- **The continuity of probability function is similar.**
- **Def.** A seq $\{E_n, n \geq 1\}$ of event of a sample space is called increasing if

$$E_1 \subseteq E_2 \subseteq E_3 \subseteq \cdots E_n \subseteq E_{n+1} \subseteq \cdots;$$

it is called decreasing if

$$E_1 \supseteq E_2 \supseteq E_3 \supseteq \cdots E_n \supseteq E_{n+1} \supseteq \cdots.$$

